

As a group, we organised our workflow around the weekly DATA3001 seminars held on Mondays from 10:00 am to 12:00 pm, using the time immediately after each seminar as our regular meeting window. This structure ensured that all group members were consistently present and allowed us to make steady progress throughout the assignment.

Our first meeting took place on **Monday, 13 October (Week 5) at 12:00 pm**. In this meeting, we clarified our understanding of the assignment requirements, discussed the dataset, and explored the expected outputs. Each group member contributed initial ideas regarding how the modelling and report-writing would be approached.

The second meeting was held after FLEXI week, on **Monday, 27 October (Week 7) at 12:00 pm**. During this session, we discussed the modelling framework we wanted to adopt. We collectively decided to avoid neural networks because they were scheduled to be taught in Weeks 9 - 10, meaning we did not yet have sufficient time or expertise to apply them rigorously in this assignment. Instead, we narrowed our focus to linear models, regularisation methods, and tree-based techniques. We also divided tasks based on our strengths:

- **Gahan** took primary responsibility for modelling and tuning,
- **Issada** focused mainly on the exploratory data analysis (EDA),
- **Parsa** worked across both modelling and EDA, with his main role being writing and structuring the report, ensuring code readability, and refining the visualisations.

Although tasks were divided, we made sure to keep all members informed and engaged with each component, so everyone understood the full workflow.

Our final meeting occurred on **Monday, 10 November (Week 9) at 12:00 pm**, where we worked together on debugging, reconciling outputs, finalising the tuned models, and preparing the final structure of the report. This was an especially productive meeting, as many final issues were resolved collaboratively. Overall, we felt the workload distribution was fair, communication was strong, and no member underperformed or contributed significantly less than the others. Across all meetings, attendance and contribution from **Parsa, Gahan, and Issada** were consistent

The most challenging aspect of the assignment was working with a relatively large dataset alongside Python packages we had limited prior experience with, particularly when handling many dummy variables and interaction terms. However, our previous exposure to Kaggle, guided examples from tutorials, and the technical foundations from DATA3001 helped us overcome these challenges. Additionally, this assignment improved our ability to work with real-world, messy datasets and to manage computational complexity in model training.

Overall, the assignment was a valuable experience in both technical and collaborative terms, strengthening our understanding of statistical learning methods while reinforcing the importance of structured group coordination and clear task allocation.

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